

RESEARCH ARTICLE OPEN ACCESS

Wind Turbine Blade Crack Detection and Assessment in Images Using Machine Learning

Callum Rothon¹  | Nina Dethlefs¹ | Robert C. Houseago² 

¹Department of Computer Science, Loughborough University, Loughborough, Leicestershire, UK | ²Department of Geography and Environment, Loughborough University, Loughborough, Leicestershire, UK

Correspondence: Callum Rothon (cocr2@lunet.lboro.ac.uk)

Received: 29 January 2026 | **Revised:** 22 June 2026 | **Accepted:** 9 July 2026

Keywords: computer vision | crack detection | crack measurement | offshore wind turbines | structural health monitoring

ABSTRACT

As the wind energy industry matures, inspection of wind turbine blades (WTBs) is shifting from a manual process involving rope access and grading of damage, towards unmanned aerial vehicle (UAV) photography and artificial intelligence-aided processing of data. Damage detection and classification methods have been proposed, using object detection based on convolutional neural networks (CNNs) to localise and quantify damage in images of WTBs. However, existing approaches are unable to measure the detected instances of damage, which is key for effective maintenance planning. A major limitation to crack measurement in images is the requirement for an explicit scale reference marker or additional data such as imaging distance, which is not always practical or cost-effective in the inspection of large WTBs. Here, we present a novel crack detection and measurement framework for WTBs which does not require additional sensor data, using a fine-tuned CNN-based object detection and segmentation network to locate cracks and implicit scale references in images. We detect WTBs within images and then identify cracks, scaling them using known dimensions of the WTBs. This work also provides a comprehensive dataset of representative WTB damage data including images of small-scale WTBs with simulated cracks, leading edge erosion and contamination. Our approach shows strong performance on the test dataset of images and can detect and measure cracks in limited orthomosaic views produced from 3D reconstructions, indicating that the approach could be applied to the inspection of full-scale WTBs in the future.

1 | Introduction

As wind energy becomes an increasingly important part of sustainability targets worldwide, a shift towards condition-based maintenance and smart operations and maintenance (O&M) is underway. This is enabled by the volume of data recorded from wind turbines (WTs) and the use of modern inspection methods such as UAV photography.

Wang et al. [1] note that WT blades (WTBs) fail more frequently than other major WT components (tower, foundation, etc.), accounting for 20% of failures in large WTs. Historically,

inspection of WTBs was conducted using rope access to the WTB surface, with operators locating and grading damage [2, 3]. More recently, UAV photography has reduced the need for rope access, reducing downtime for inspection [3].

Significant progress has been achieved in applying machine learning (ML) and deep learning (DL) models to O&M in the wind energy industry [4, 5], including inspection of WTs [6–10] and prediction of faults and failures [11].

Before the adoption of ML methods, processing UAV-gathered images was a manual process that was time and

Abbreviations: AI, artificial intelligence; CNN, convolutional neural network; CV, computer vision; DL, deep learning; GFRP, glass fibre reinforced polymer; ML, machine learning; OSW, offshore wind; O&M, operations and maintenance; SHM, structural health monitoring; WT, wind turbine; WTB, wind turbine blade.

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2026 The Author(s). *Wind Energy* published by John Wiley & Sons Ltd.

resource-intensive and vulnerable to human error. DL methods have been successfully applied to damage detection in images of WTBs [6, 7, 12–16], assisting staff by locating and classifying damage in images, reducing the time taken to inspect a turbine.

Efficient O&M planning requires knowledge of the state of an asset and the timescale for action. Before the remaining useful life (RUL) can be estimated, the current state of the WTB and any damage present must be quantified. The majority of current ML crack detection tools for WTBs are not capable of measuring the dimensions of predicted cracks in real units [6, 7, 9, 10, 12–15] which we consider a major limitation.

To improve O&M planning in the wind energy industry it is necessary to implement a pipeline for measurement of cracks on WTBs which can integrate with current inspection procedures. In this work, we present a method for crack detection and measurement on WTBs, based around an ML crack detection network. We train and validate our method on a novel dataset of images of damaged WTBs, which we make available. In the following sections, we review current literature on image-based damage detection on WTBs and crack detection (Section 1.1), describe how we produced the damaged WTB dataset (Section 2), present (Section 3) and validate (Section 4) our crack measurement approach, and discuss our findings (Section 5).

1.1 | Related Work

The detection and measurement of cracks in images is a long-standing area of research in the field of computer vision (CV). Early methods were based on classical image processing, using filters such as skeleton, pruning and edge detection [17]. These methods remained relevant up to the rise of DL methods for image processing, with Zhang et al. [18] finding that decision trees outperformed contemporaneous ML methods, including support vector machines and neural networks.

CNN-based object detection networks have been applied to damage detection in images of WTBs, including in UAV-gathered RGB images [9, 10, 13, 14, 16, 19–21] and thermographic images [6, 7]. Although these approaches have shown good performance in the detection of damage, including cracks, most approaches do not measure or quantify the damage detected, which we consider a key limitation and the logical next step from detection. Li et al. [10] proposed a damage detection and measurement method based on SegFormer [22], reporting good results in the detection task, achieving an F1 score of 98.52%. Their measurement method calculates a damage ratio based on the area of the detected damage and the area of the image. A similar method is proposed by Hang et al. [9], which proposes a damage index based on the relative length and width of the crack in pixels. However, neither method gives a measurement in real units (e.g., mm), which limits their usefulness as a basis for O&M scheduling.

Due to the limited work identified on crack measurement on WTBs, related work from other fields was researched. Zhang et al. [18] noted that most contemporary literature on crack detection focused on crack detection in concrete structures, which appears to still be the case in more recent publications. Since

Zhang et al. [18] published their paper, multiple CNN-based approaches have been successfully applied to crack detection and measurement in images of concrete and masonry structures, showing strong results [23–27].

Yuan et al. [23], Minh Dang et al. [24], Nyathi et al. [25], Li et al. [26] and Giulietti et al. [27] have all reported excellent results in crack detection tasks in concrete structures, with Giulietti et al. [27] reporting a near-perfect F1 score of 99.9%. Nyathi et al. [25] extend crack detection and classification to segmentation of cracks, achieving 99.2% and 96.5% accuracy in classification and segmentation tasks. Although these results are impressive, the images of concrete structures are simple compared with UAV-gathered images of WTBs, with cracks taking up much of the image.

Although the detection of cracks in images can be achieved without additional input, a key limitation of current crack measurement methods is their reliance on additional information for scaling of cracks in images. Methods can be grouped into those that use an explicit scale reference marker in the image [23, 24] and those which use additional data, such as imaging distance [25, 27].

Good results were achieved using scale reference makers by Yuan et al. [23], who combine a CNN crack detector with image processing techniques to locate the crack tip and measure the crack. Their method achieves a measurement error of 0.8%–1.6%. Positioning a scale reference marker in the image is suitable in some use cases, where the marker will be clearly visible and can be placed easily, but is impractical in the inspection of large and inaccessible structures. The marker must also be positioned directly in plane with the cracks and photographed from an orthogonal angle, or distortion will occur and the scaling will be inaccurate.

A related approach is the use of an object of known dimensions in the image as the scale reference. Minh Dang et al. [24] propose a method to measure cracks in masonry using the dimensions of bricks, using DL models to detect bricks and cracks. This method achieved measurement error of 10%, which is encouraging given the lack of an explicit reference marker, but is far greater than the error reported by Yuan et al.

Reference-free approaches have been developed, based on measurement of distance between the camera and surface during imaging. Nyathi et al. [25] and Giulietti et al. [27] use laser and infra-red depth sensing, respectively, to calibrate the crack measurement. Nyathi et al. [25] present a method based on DL and image processing for classification, segmentation and measurement of crack width, achieving mean average error of 0.16 mm on cracks with widths of 2–9 mm. Li et al. [26] report relatively small errors for crack measurement of between 0.3% and 2.8%, comparable to results achieved by Yuan et al. [23]. Giulietti et al. [27] report measurement error below the lateral resolution of the laser profile scanner used, indicating that error present could reasonably be from the data acquisition process. Although these methods prove successful and attain low error, depth sensing adds additional cost and complexity to inspections, and the handling of this additional data adds to the computational cost of crack measurement.

It must be noted that the properties of concrete structures differ significantly from those of WTBs, which consist of a glass fibre reinforced polymer (GFRP) shell with wood and foam supports. Therefore, different challenges arise during imaging and processing. Crack detection on concrete has generally focused on orthogonal views of flat surfaces where cracks are in plane. UAV images of WTBs are more complex, with curved geometries, so cracks are not in a flat plane and image views are often not orthogonal. The curved geometries of the blade also make maintaining consistent imaging distances more difficult than on a flat surface. These differences will impact how methods from concrete crack detection can be applied to WTBs.

When discussing UAV inspection of WTBs, Sterckx et al. [28] note that although 2D images are generally sufficient to detect damage, 3D reconstructions enhance the quantification of damage as they provide detailed surface topography. 3D reconstructions of damaged WTBs enable more insightful measurement of damaged areas than 2D images alone, including damage dimensions and depth profiling through analysis of the texture and surface roughness.

Photogrammetry techniques have been applied to a range of tasks in the wind energy industry, including inspection of damaged WTBs [28, 29], and to the inspection of photovoltaic (PV) energy infrastructure [30]. Oliveira et al. [30] use Mask R-CNN to segment faults in thermal images of a PV farm, then reconstruct hybrid views with highlighted faults at the PV farm level. Similarly, object detection has been performed in 3D reconstructions by Lu et al. [31], using a modified YOLOv5 network to detect vehicles in reconstructed views generated from UAV images.

Although UAV-aided inspection of WTs is an active area of research, few datasets have been made available in the public domain. A leading example is the dataset published by Shihavuddin and Chen [32], referred to as the 'DTU dataset', which contains 701 UAV-gathered images of a WT from two inspections and was used in the authors work on WT damage detection [20]. Another example is the Blade 30 dataset published by Yang et al. [33], who compiled 627 images from 30 separate sets of images of onshore and offshore WTBs, including examples of cracks, LEE and contamination.

A major limitation of these datasets for application in analysis of WTB damage is that they do not contain ground truth dimensions for instances of damage, limiting their use when validating a measurement tool. We investigated two methods to adapt the datasets for damage measurement: the use of known dimensions within the image to scale instances of damage, and the use of 3D reconstruction. However, both methods proved unsuitable: scaling due to a lack of scale reference objects that could be used to estimate the dimensions of instances of damage, and 3D reconstruction due to insufficient views of the WTBs, leading to failed reconstructions. Although the DTU dataset [32] contains views around the WT, the leading and trailing edges are under-represented, leaving gaps in the reconstruction at critical areas for crack formation. The Blade 30 dataset is intended for use with the image-stitching approach proposed by Yang et al. [33] and contains views of sections along the WTBs. However,

the views only contain one face of the WTB, and are therefore not sufficient for building a 3D reconstruction. Therefore, these datasets were proven unsuitable for the training and validation of a damage detection and measurement tool, necessitating the creation of a suitable dataset containing sufficient views for reconstruction and defects of known dimensions.

1.2 | Research Gaps

After reviewing related work on crack detection, measurement and ML-aided WTB inspection, we identified multiple gaps in the literature around crack measurement in UAV-gathered images of WTBs. The lack of high quality data depicting WTBs with cracks of known dimensions significantly limits work in this field, as crack measurement methods cannot be validated without suitable data for detection and measurement tasks. The majority of current literature on crack detection and measurement focuses on inspection of concrete structures. Although the overall approaches may be transferable to WTBs, the differing materials and geometries limit application of existing methods to WTBs. In the field of measurement within images, the requirement for an explicit scale reference or additional data such as imaging distance limits where existing methods can be applied. If scale and accessibility are not issues, it is trivial to add a scale marker to an image or to control imaging conditions to a point where camera intrinsics can be used for scaling. However, in applications where access is limited and large-scale assets are the subject of the inspection (e.g., large WTBs), it is impractical to position markers in each image. If the dimensions of components are known, then these may be used to provide scale references, provided they can be reliably detected. Without robust measurement, the assessment of cracks is limited to the judgements of O&M staff. This prevents further optimisation of O&M planning, as the RUL cannot be estimated without detailed knowledge of the current state of the asset.

1.3 | Contributions

In this work, we develop a framework for DL-based crack detection and measurement on WTBs, addressing the differing requirements from existing methods for inspection of concrete structures. Our proposed method reduces the need for time-intensive review of images and advances capabilities to assess the state of WTBs over existing detection and classification networks by measuring detected cracks, without the requirement for additional input or sensing equipment. To validate our approach, we generate a dataset of images depicting damaged small-scale WTBs, with measured crack lengths and sufficient views for high quality 3D reconstruction. These images simulate the views seen in existing datasets while overcoming their limitations for measurement and reconstruction tasks.

We make the following contributions:

- We produce and present a novel dataset of images of damaged small-scale WTBs, suitable for 3D reconstruction, crack detection and measurement without requiring camera intrinsics or distance measurement.

- We propose a method for measurement of cracks in images of WTBs, using CNN-based detectors to locate cracks and scale references in images and calculate crack length by scaling from known dimensions of scale reference objects.
- We validate the proposed crack detection and measurement pipeline which we demonstrate using our dataset, trialling on both 2D photographs and views of 3D reconstructions.

2 | Materials

As the WTB damage datasets currently available in the public domain [32, 33] were found to be unsuitable for crack measurement through current methods, we produced a novel dataset to train and validate our WTB crack measurement pipeline, considering the requirements for views and crack data and UAV inspection procedures.

2.1 | WTB Damage

Before generating our dataset, we researched damaged WTB datasets to characterise the appearance of damage. Common forms of damage found in large WTBs and current methods of damage detection are reviewed by Wang et al. [1].

Cracks form in areas of the WTB which undergo high cyclic stresses—the root, tip and trailing edge. Mild cracks may be invisible to the naked eye and below the resolution of cameras at the range of a UAV inspection. More severe cracks present as jagged black or grey lines against the light coloured coating of the WTB.

Delaminations are a related form of damage, where the matrix between layers of glass fibre weave fails. Delaminations can occur at lower stresses than cracks due to weakness at adhesive joints. Visually, delaminations appear similar to cracks and generally occur at the trailing edge of the WTB.

Lightning damage causes characteristic burns around the point of contact at the blade tip. In severe cases, lightning

damage can cause major cracks and delaminations to form instantaneously as intense heat causes the GFRP structure to fail.

Leading edge erosion (LEE) occurs where suspended particles in the air such as dust, dirt and water droplets gradually wear away the coating at the leading edge. In mild cases, LEE presents as roughness and pitting on the WTB surface, whereas severe LEE will expose the underlying GFRP through the coating.

Contamination from dust, dirt, rust streaks and bird strikes can also produce the appearance of damage on the WTB surface. Differentiating between contamination and LEE can be difficult due to the similar appearances.

Examples of damage from the Blade 30 dataset [33] can be seen in Figure 1.

2.2 | Damaged Small-Scale WTBs

To overcome the limitations of available WTB datasets, we produced set of images of synthetically damaged WTBs under controlled conditions, simulating the views captured in UAV inspections of large-scale WTs, such as the Nordtank DTU [32] and Blade 30 [33] datasets. As surveying a large-scale WTB was impractical, we opted to photograph small-scale WTBs from consumer WTs with varying amounts of damage. We were able to control all elements of the dataset, including damage types, severity and photographing conditions including lighting and coverage. Therefore, we were able to overcome two major obstacles: the lack of known crack lengths for validation of our measurement pipeline, and the lack of sufficient coverage for 3D reconstruction.

2.2.1 | Blade Types

The dataset includes damaged and undamaged views of three types of WTB, shown in Figure 2 and described in Table 1. These blades are replacement parts for domestic WTs, with rotor diameters of 0.8–1.2 m. The WTBs have curved aerofoil surfaces

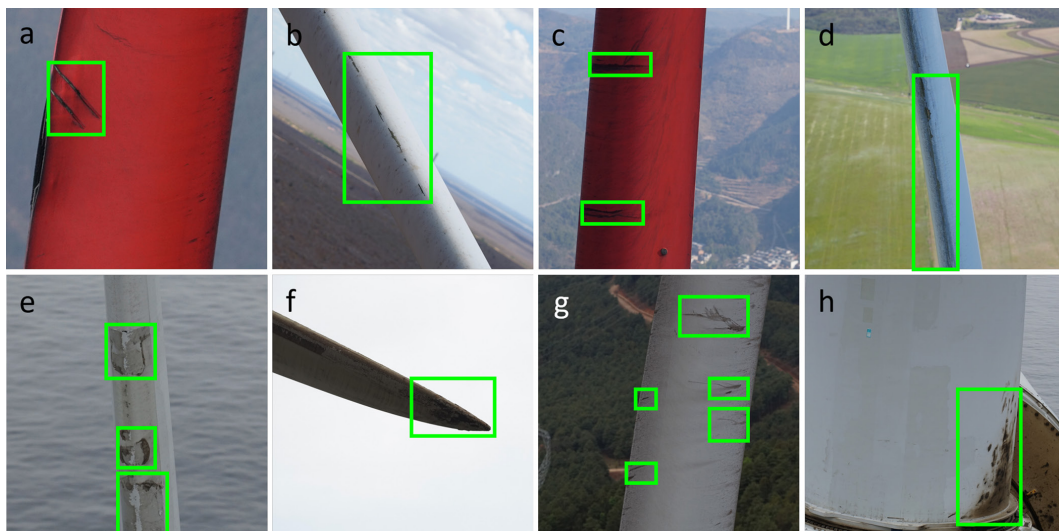


FIGURE 1 | Examples of WTB damage from the Blade 30 dataset [33], including examples of cracks (a–d), LEE (e, f) and contamination (g, h).

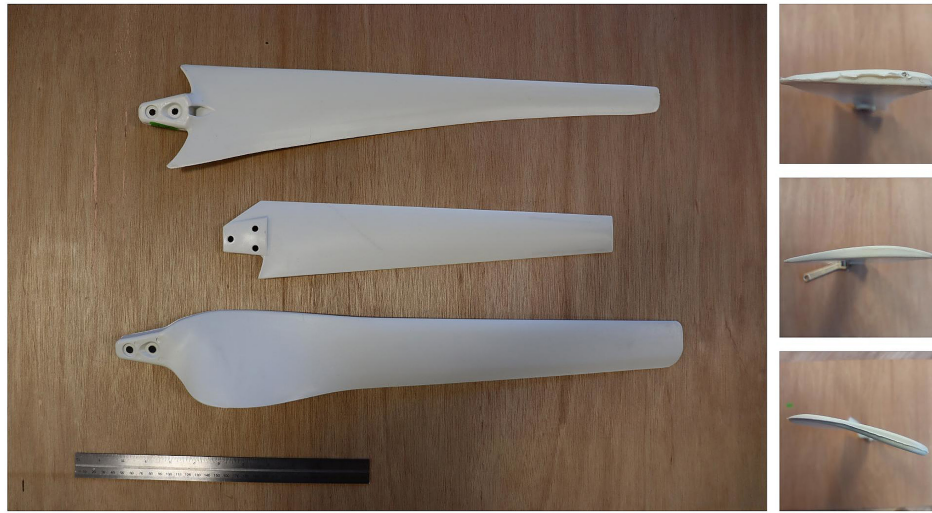


FIGURE 2 | Small-scale WTBs with a 30.5 cm steel rule for scale and cross sections: Type 1 (top), Type 2 (middle), Type 3 (bottom).

TABLE 1 | Small-scale WTB model, key dimensions (length, width) and usage in the ML framework (train, validation, test).

WTB type	Model	Length (mm)	Width (mm)	Usage
Type 1	Meteor MZ-3000	550	115	Train/Val/Test
Type 2	Meteor MK-8B	400	80	Train/Val/Test
Type 3	Meteor M6-6000	580	96	Fine-tune/Val/Test

and a white finish similar to large-scale WTBs, making them suitable for demonstrating the potential of the proposed crack length measurement framework before the process can be extended to large-scale WTBs.

These WTBs feature different dimensions and geometries, and we intend to use them to investigate how well a trained network will be able to generalise to previously unseen WTB types with minimal additional training, described in detail in Section 3.2.4.

2.2.2 | Damage

For the purposes of this study, it was considered that simulated damage which resembles real cracks, LEE and contamination on WTBs would be sufficient to evaluate the performance of the proposed crack detection and measurement pipeline, without simulating the mechanisms which drive the formation of cracks in full-scale WTBs. To simulate cracks, a scribe was used to create a scratch in the surface of the WTB, which was highlighted with ink to replicate the colour difference seen in cracks on large WTBs. Crack lengths started at 10 mm and were then extended in steps of 5 mm up to a length of 25 mm, measured to a precision of 0.5 mm using a steel rule. LEE was simulated by creating

wear on the leading edge of the WTB using sandpaper and highlighted using ink to replicate the real world appearance of LEE. Contamination on the WTB surface was applied using water-based ink, introducing blemishes and smudges to the WTB surface. In this paper, we do not aim to detect or quantify LEE and contamination, but we note that this could be achieved in future using this dataset.

A detailed view of a simulated crack with simulated LEE and contamination is shown in Figure 3, compared with an example of a crack from the Blade 30 dataset [33].

2.3 | Image Acquisition

2.3.1 | Views

The dataset contains images from a range of angles, intended to be used to train the model for a range of photographing scenarios. The flat views are intended for use when testing the crack measurement pipeline. The UAV-style views were used to build 3D reconstructions and generate orthomosaic (OM) views of the small-scale WTBs, with the WTB scaling crack measurement pipeline also being trialled on the OM views.

The WTBs were positioned on a flat plywood board, with two bright green target markers of known dimensions (30 mm square) placed diagonally around the WTB to allow scaling using an explicit scale reference, which serves as a benchmark to compare our WTB scaling approach to.

2.3.1.1 | Flat Views. Flat views were taken of the WTBs at each crack length and with varying levels of contamination and LEE. Example 2D views of undamaged and damaged WTBs are shown in Figure 4. For each state (undamaged, crack lengths, LEE and contamination) views of both the pressure side (front, facing the wind) and suction side (rear) of the WTB were taken. The orthogonal view and markers in plane with the WTB make the flat views similar to the images used in crack detection in concrete, allowing more direct comparison between methods.



FIGURE 3 | (L) Detailed view of a 25 mm simulated crack close to the trailing edge of a Type 1 small-scale turbine WTB, with simulated contamination and LEE. (R) Cropped example of a crack from the Blade 30 dataset [33].

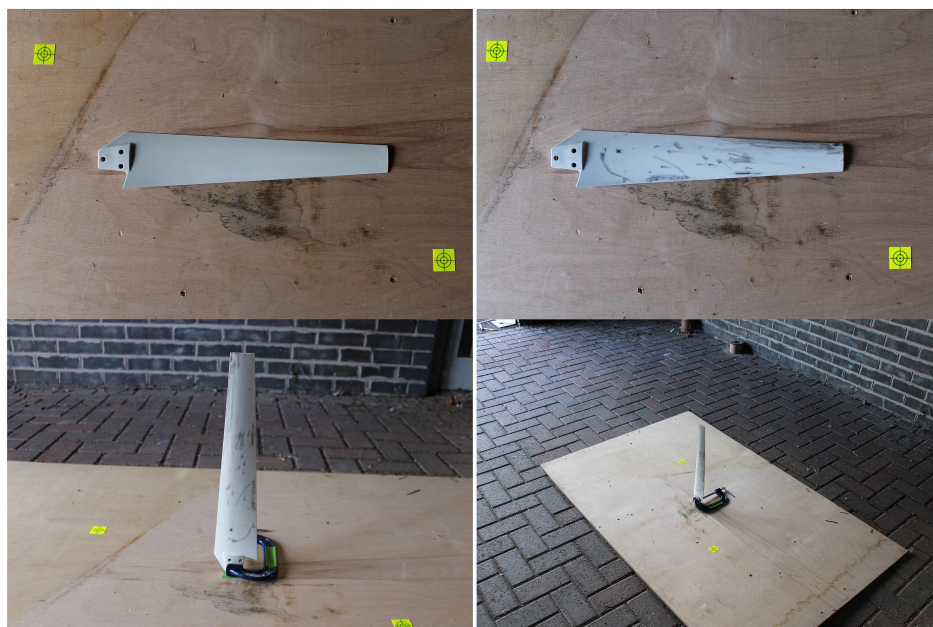


FIGURE 4 | Example views of a Type 2 small-scale WTB positioned on a plywood board with markers visible at the top-left and bottom-right corners: undamaged flat view (top left), flat view with a crack and contamination (top right), full WTB view (bottom left), global view (bottom right).

2.3.1.2 | UAV-Style Views. A set of UAV-style views was captured, supporting reconstruction into a 3D model in Metashape to build a full model of the damaged WTB (detailed in Section 3.1). These views were intended to simulate real-world UAV inspection, giving full coverage of the WTB with overlap between images in line with recommendations for 3D reconstruction through structure from motion (SfM). The WTB was mounted upright in a G-clamp, allowing both pressure and suction sides to be photographed. The UAV-style views consisted of two types:

- **Full-WTB views:** These views focused on acquiring close up imagery of the WTBs from a range of angles: approximately 90°, 60°, 30°, and from above, circling around the WTB to build up around 72 views of the WTB with full coverage. The images were taken so that at least one scale marker was present in each image.
- **Global views:** These views were taken at a greater distance to add context of the surrounding scene to improve

3D reconstruction quality. Eight views were taken around the model at approximately 45° angles.

Examples of full-WTB and global views are included in Figure 4. Full sets of views (top down, full WTB views, global views) were taken at each stage of damage, totalling 82 views per stage. The full WTB and global views were used to reconstruct a 3D reconstruction and OMs in each damage condition, as described in Section 3.1.

2.3.2 | Photographing Conditions

In line with recommendations from Woelk and Sier [34], the small-scale WTBs were photographed under natural outdoor lighting to reduce issues with shadows, reflections and colour, which could impact the quality of the reconstructions. These measures aimed to create conditions which would result in images suitable for both crack detection and 3D reconstruction.

3 | Methods

To determine the length of detected cracks, we propose to use the known dimensions of the WTB for scale, overcoming the need for an explicit scale reference in the image which limits the application of existing approaches to WTBs. We use the WTB dimensions and predicted bounding box coordinates to calculate the mm/pixel scale at the WTB surface and then convert the crack coordinates from pixels to a crack length in mm. We also trial our crack measurement pipeline on OMs generated from the 3D views of the small-scale WTBs, to determine if our approach can be applied to hybrid views.

To set a benchmark for our proposed approach, we built a pipeline which used the reference markers to calculate the mm/pixel scale. This allows better comparison to existing approaches which rely on the use of an explicit scale reference in the image, as we can compare the performance of both approaches on the same dataset. Crack measurements were also calculated from the ground truth annotations to quantify the level of error present from the data preparation.

We train three separate detection networks for the cracks, WTBs and markers, which predict segmentation masks and bounding box coordinates. When properly trained, the bounding boxes will tightly bound the objects and cracks, allowing a robust measurement of crack length.

3.1 | 3D Reconstruction

3.1.1 | Reconstruction Methods

Photogrammetry is a family of image processing techniques which aim to produce accurate 3D models from a set of photographs of an object or scene. When scaled, these models can be used for 3D dimensional analysis of scenes which contain larger regions than a single photograph. SfM [35] is a subset of photogrammetry methods, where the structure of an object or scene is calculated from multiple views taken from different positions, such as frames from a UAV flight. SfM extends the principles of stereo vision, where corresponding features between multiple images are leveraged to build a 3D model from a set of 2D images, without requiring additional information such as camera positions or distance measurement.

The workflow for converting a set of 2D images into a 3D model through SfM is summarised by Li et al. [36]. The major steps involve extracting and matching features between images to estimate camera positions, then triangulating and refining the positions of points in 3D space. A sparse point cloud containing relatively few points is constructed to show the coarse structure, before a dense point cloud with more points is generated, showing fine detail.

OMs are images generated by ortho-correcting and combining multiple overlapping views of a scene or object [37, 38]. They are commonly generated from aerial photos gathered by UAVs and have been applied in a range of fields, including environmental monitoring and inspection of infrastructure.

3.1.2 | 3D Reconstruction Process

3D reconstructions were generated using Agisoft Metashape [39], which converts 2D photographs into 3D models through photogrammetry and allows point clouds and OM images to be generated. Metashape has been applied to a wide range of applications, including processing UAV photographs of WTBs [29].

Woelk and Sier [34] provide guidelines to generate high-quality 3D reconstructions from UAV-gathered images. Images must be of high quality, with good lighting and minimal noise. Ground sampling distance (GSD), the distance between the centres of adjacent pixels at ground level, should be minimised to give the highest resolution possible and kept consistent. Image overlaps of 60%–90% are advised depending on feature density.

Our procedure for reconstructing images into a point cloud and generating OMs followed the process as described by Li et al. [36] and is as follows:

1. **Filter low quality images:** The quality of all images was estimated, and images with low quality were removed from further analysis if blurring was found on the WTBs. In many cases, low quality was due to the background being out of focus, whereas the WTBs were in focus.
2. **Align photos (sparse point cloud):** The relative positions of camera views were determined by matching features between images, which results in a sparse point cloud.
3. **Cropping:** After construction of the sparse point cloud, the scene was cropped to the area surrounding the WTB and markers to reduce size of the model and remove the background of the scene.
4. **Build dense point cloud:** The point cloud was constructed from the aligned photos. When set to 'high' quality, this step took 1–2 h with GPU acceleration.
5. **Generate OMs:** Finally, we generated OMs of chosen views of the WTB. After trialling a range of scales, a maximum dimension of 4096 pixels was applied, building from the dense point cloud, as this was found to give sufficient detail without excessive processing.

The parameters used for the reconstructions are included in Table 2.

The OM shown in Figure 5 is an example of an OM considered as suitable for crack detection, with the crack, LEE and

TABLE 2 | Metashape reconstruction parameters.

Parameter	Value
Alignment accuracy	High
Point cloud quality	High
Point cloud filtering	Aggressive
Orthomosaic resolution	0.1 mm/pixel
Orthomosaic max. dimension	4096 pixels

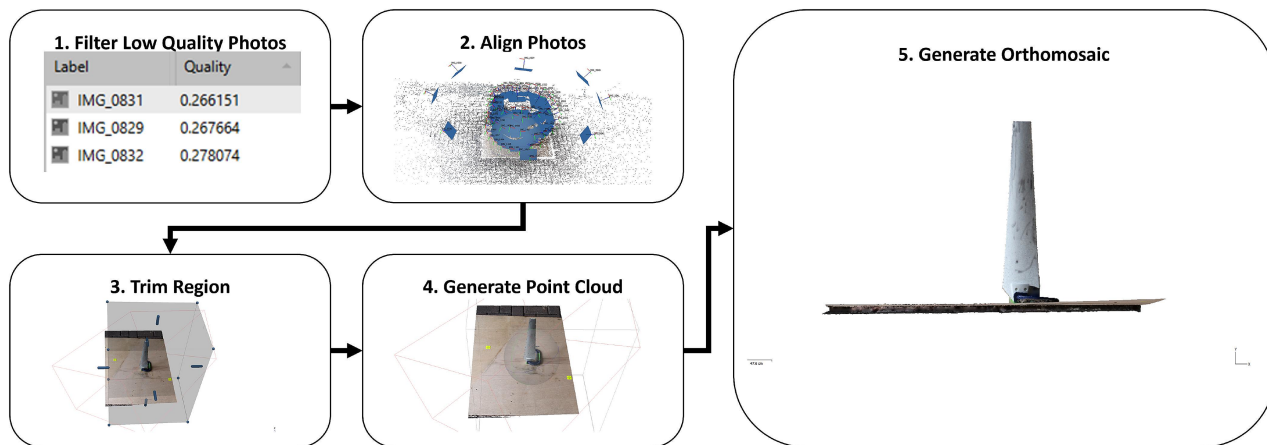


FIGURE 5 | Metashape reconstruction outputs at each major stage.

contamination visible, and without noticeable distortion, gaps, seam lines or shadowing.

3.2 | Damage Detection Network

3.2.1 | Damage Detection in Images

Object detection and segmentation in images is a widely researched field, with popular ML object detection approaches including the Region-CNN (R-CNN) family [40–43] and You Only Look Once (YOLO) family [44]. These object detectors use CNNs to extract features from images and predict the location and class of predicted objects.

The R-CNN family [40–43] uses a two-stage architecture, where regions of interest (RoIs) which contain an object are proposed, and then a classification for the object contained in the RoI is predicted. The original R-CNN [40] ran slowly as each object proposal was separately passed through the CNN for feature extraction and classification. Fast R-CNN [41] improved speed and performance by extracting features from the entire image using the CNN before the RoI proposal, and Faster R-CNN [42] built on this further by implementing a region proposal network (RPN) to propose RoIs, which share convolutional layers with the R-CNN detection model. Mask R-CNN [43] extends object detection into segmentation and outputs bounding box coordinates, classifications and segmentation masks which highlight pixels contributing to the prediction. The YOLO family uses a single-stage architecture where detection and classification are trained as a single task [44], reducing runtime during training and inference.

Object detectors are commonly applied to detection of everyday objects in tasks such as ImageNet [45] and Common Objects in Context [46], with pre-trained object detection weights available for common models including Mask R-CNN and Faster R-CNN. For closely related tasks, sufficient performance may be attained by using pre-trained weights without additional training, known as generalisation. In many detection tasks, the deep features extracted by the CNN backbone network will be similar, so the weights of a detection network can be fine-tuned to a new task with a small amount of training data [47], known as few-shot learning (FSL) [48]. These methodologies

fall under the umbrella of transfer learning (TL), where models proven successful in one application are transferred to a related task, reducing the requirements for training data, and time and resource-intensive training [49]. As collection and preparation of UAV images is a time and resource intensive process, it is beneficial to minimise the amount of data that is required to train a model to a high standard.

3.2.2 | Data Preparation

Annotations for training the detection models were generated using the VGG Image Annotator (VIA), developed by the Visual Geometry Group [50]. The annotations from VIA were converted into binary segmentation masks, which were used to calculate bounding box coordinates for each instance of object (WTB, marker) or crack during training.

Images containing WTB Types 1 and 2 were used to train the three detection models, and images containing WTB Type 3 were reserved to test TL to a previously unseen WTB type with limited data. The annotations were also used to benchmark the level of error in the dataset from the annotation process, by calculating crack length directly from the ground truth bounding boxes. This allows us to compare the performance of our model to a human reviewing the images.

3.2.3 | Training

For the detection of cracks and scale reference objects, we apply a set of Mask R-CNN [43] models, with separate models trained for the detection of markers, WTBs and cracks. Mask R-CNN forms the basis of the methods proposed by Zhang et al. [21] and Wang et al. [13] for damage detection on WTBs, showing promising results. We use a ResNet 50v2 backbone network for feature extraction, as this configuration showed good performance on a WTB damage detection task when applied by Roth et al. [16]. The structure of Mask R-CNN is shown in Figure 6, showing an example of WTB detection in an image.

Initially, a single model was trained to detect cracks, markers and WTBs. It was found that this network struggled to propose RoIs for cracks, due to their small scale compared with the

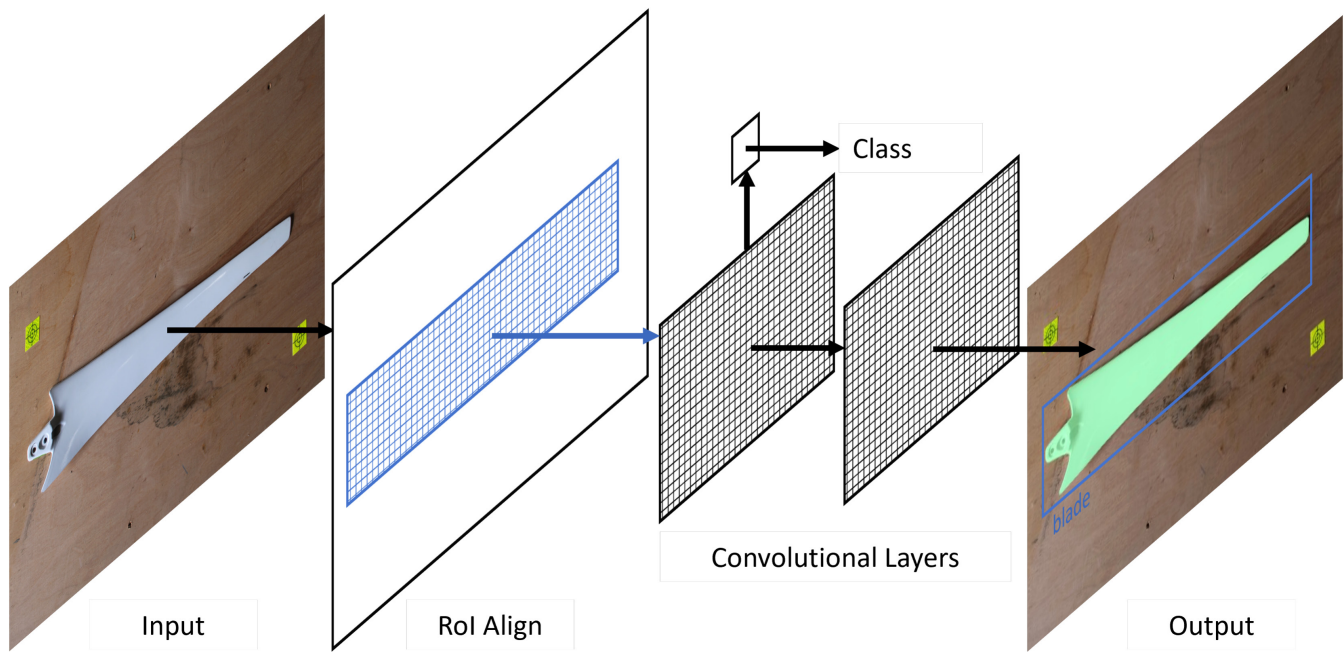


FIGURE 6 | Example Mask R-CNN process, showing an input image, region proposal using RoI Align, class prediction, convolution and outputs (bounding box, segmentation mask, class prediction).

markers and WTBs. This was overcome using a parallel detector approach, where the detection of WTBs, markers and cracks was handled by dedicated networks trained independently of each other, allowing the models to specialise to their respective tasks.

The models were initialised using pre-trained weights for the COCO object detection task to reduce training requirements. The pre-trained Mask R-CNN models were fine-tuned to our damaged small-scale WTB dataset, using an 80:20 train:validation split of images of WTB Types 1 and 2. Both 2D and 3D images were compiled into the training and validation datasets for the marker, WTB and crack detection models, allowing the models to be used for both flat views and OMs output from the 3D reconstruction.

The fine tuning process is shown in Figure 7, and the training parameters are included in Table 3.

It was found that the separate networks trained for detection of markers, WTBs and cracks showed good performance in their respective tasks after 15 epochs, with the WTB and marker detection networks achieving virtually perfect P, R and F1 scores and the crack detection model showing fair performance. Training and validation loss curves for the crack detection model are plotted in Figure 8, showing convergence by 15 epochs.

Once trained to convergence, bounding boxes were well-regressed, making the parallel detector approach suitable for crack measurement. Examples of blade, marker and crack bounding boxes are shown in Figure 12.

3.2.4 | TL

Following training and validation on WTB Types 1 and 2, the models were trialled on Type 3 WTBs to examine how the model

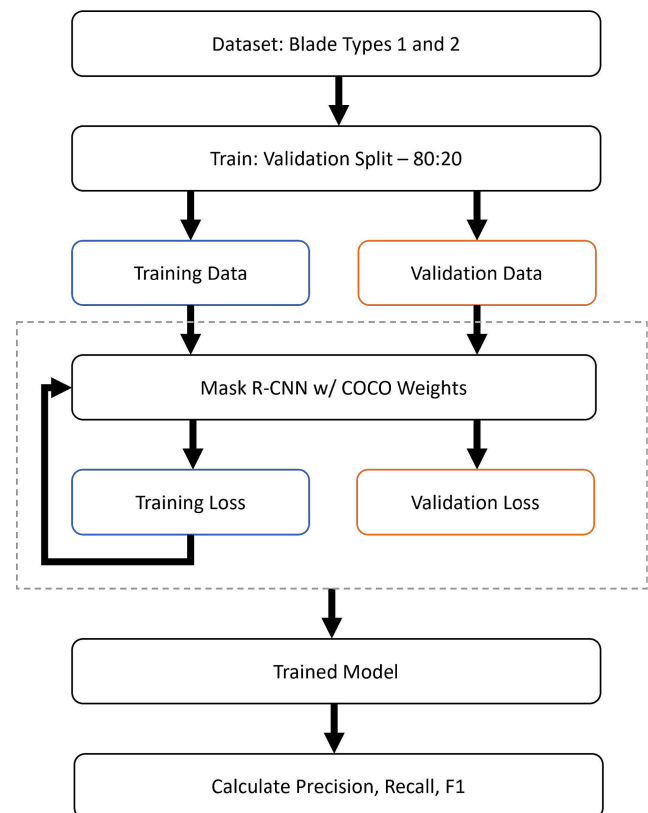


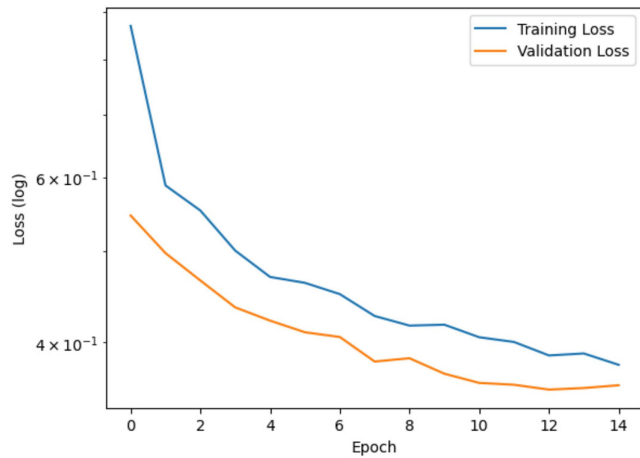
FIGURE 7 | Mask R-CNN model fine-tuning process.

would generalise to unseen WTBs, and how TL may impact performance.

Train and validation datasets were randomly sampled from the images of WTB Type 3, using a 20:80 train:validation split,

TABLE 3 | Mask R-CNN model training parameters.

Parameter	Value
Pre-trained weights	MS COCO
Optimiser	AdamW
Learning rate	1e-5
Epochs	15
Image size	3000 x 2000
Batch size	4

**FIGURE 8** | Training and validation loss curves for the crack detection Mask R-CNN model over 15 epochs.

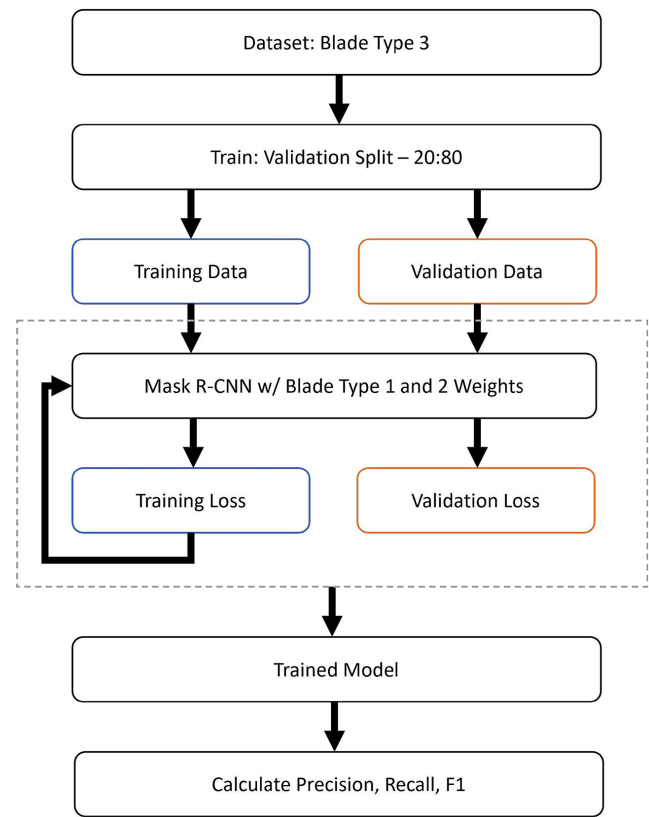
resulting in a training set of 156 images for fine tuning and a validation set of 1408 images.

The blade, marker and crack detection models were trialled on the unseen validation images of Type 3 WTBs without fine-tuning, with the blade and marker models showing near-perfect performance, but the crack detection model showing lower performance than on WTB Types 1 and 2.

As the blade and marker detection models showed good performance, no fine-tuning was deemed necessary. The crack model was fine-tuned on the small training set of images of WTB Type 3 in a TL task. The existing crack detection model trained on WTB Types 1 and 2 was loaded and trained for a further 15 epochs, resulting in improved performance. The process for fine-tuning the crack detection model to WTB Type 3 is shown in Figure 9.

3.2.5 | Blade Region-of-Interest Cropping

To prevent the image background from impacting the recognition of WTBs, Sheiati et al. [8] propose a method to segment WTBs in images. We apply a similar method to filter predicted cracks in the images, discarding crack predictions which are outside of the blade bounding box as predicted by the blade detection model, treating the blade as an RoI. Examples of WTB RoIs are shown in Figures 10 and 11 in the context of the crack measurement pipelines.

**FIGURE 9** | Mask R-CNN WTB Type 3 fine-tuning process.

3.3 | Crack Measurement

The trained marker, WTB and crack detection models were then ready to be applied to crack length measurement. As the dimensions of the scale markers and WTBs were known, they could be used to calculate a mm/pixel scale from the predicted bounding box coordinates, which could be used to convert crack bounding box coordinates from pixels to mm.

3.3.1 | Crack Scaling

For the scaling dimensions, we use the diagonal dimension of the bounding boxes. This was selected, as cracks are thin, and are unlikely to be oriented perfectly horizontally or vertically. Therefore, when bounding boxes are well regressed, the ends of the crack should be at opposite corners. The diagonal dimension is defined as the distance between the top left (x_1, y_1) and bottom right (x_2, y_2) corners of the bounding boxes. The diagonal dimension was calculated as follows:

$$d = \sqrt{\text{width}^2 + \text{height}^2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (1)$$

The mm/pixel scale can then be calculated from the bounding box diagonal dimension and the known dimensions of the marker or WTB as follows:

$$\text{scale} = \frac{D}{d} = \frac{\sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2}}{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}} \quad (2)$$

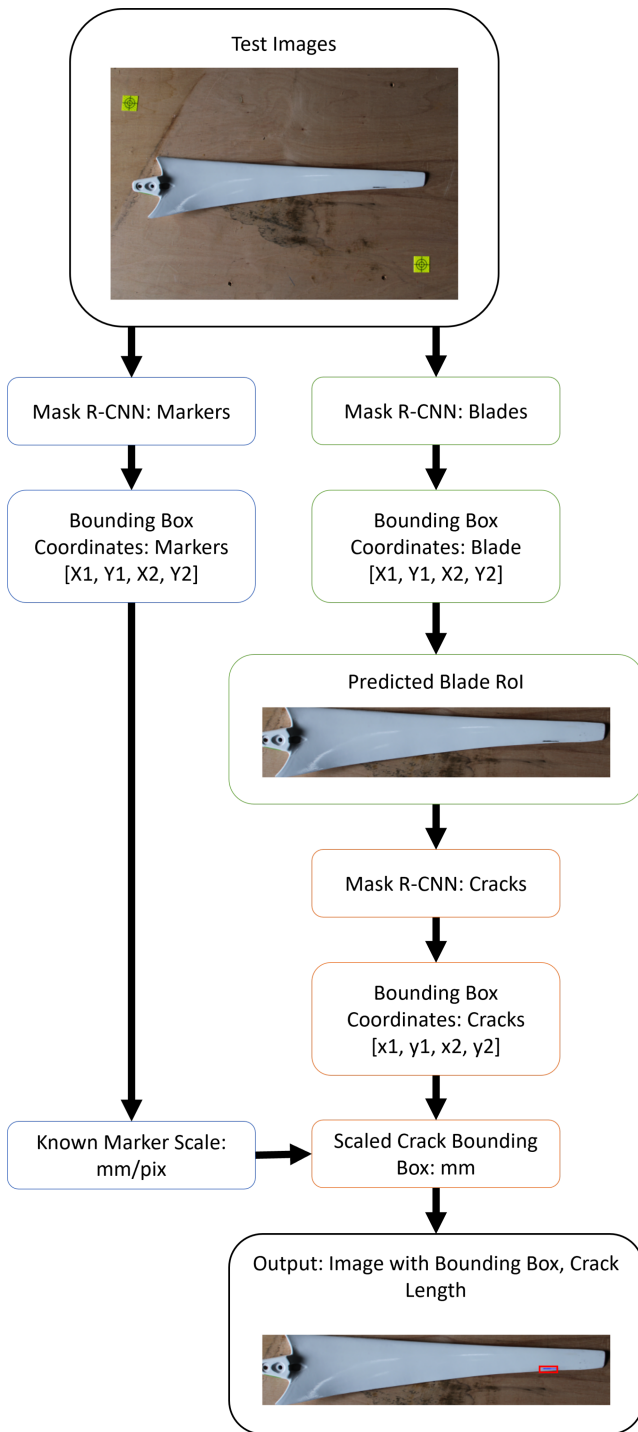


FIGURE 10 | RoI crop inference flowchart—marker scale reference.

where upper-case dimensions (D, X, Y) correspond to known dimensions of the scale reference object (marker, WTB), and lower-case dimensions (d, x, y) correspond to predicted bounding box dimensions. The WTB measurements were based on the length from root to tip and the width at the widest point of the WTB.

The pixel coordinates of the predicted crack bounding box can then be converted to crack length in mm by multiplying the diagonal dimension of the crack bounding box by the scale, which can be represented as follows:

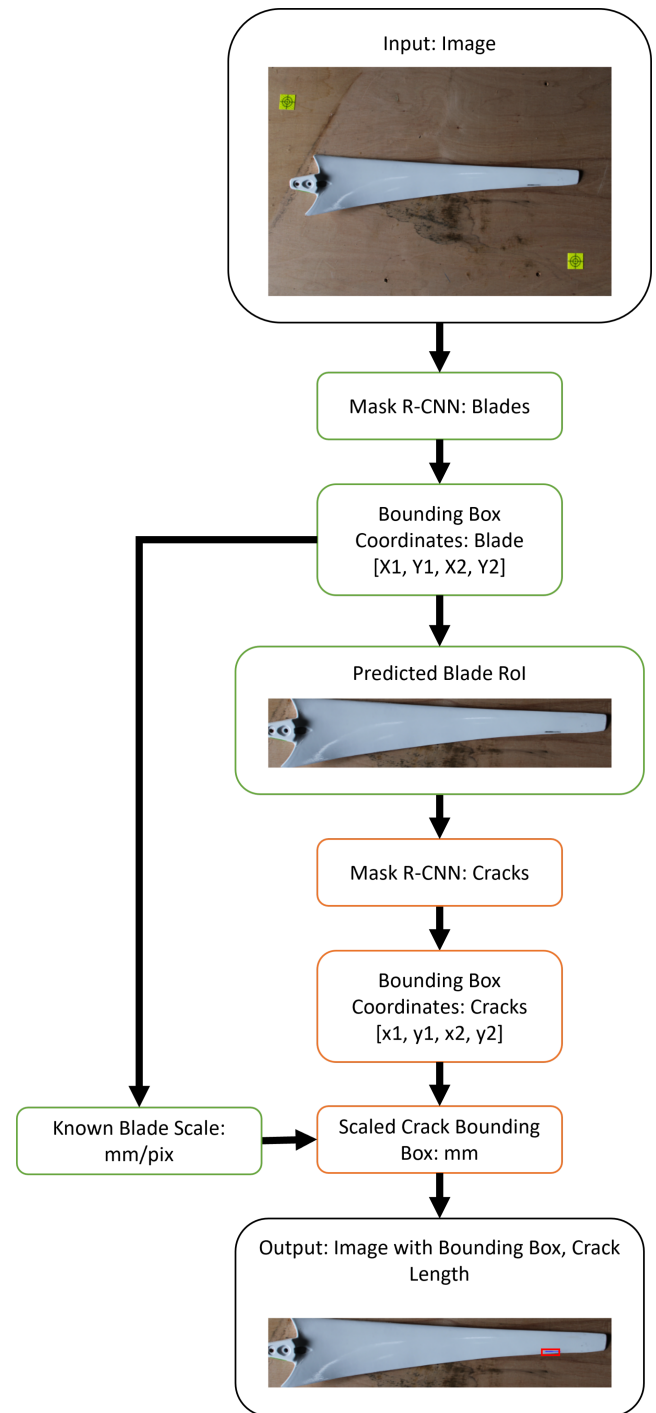


FIGURE 11 | RoI crop inference flowchart—WTB scale reference.

$$crack\ length = scale \times d \quad (3)$$

where d corresponds to the diagonal dimension of the predicted crack bounding box.

3.3.2 | Crack Detection and Scaling Pipeline

With Mask R-CNN models for detection of WTBs, markers and cracks trained, and a methodology devised to scale predicted cracks based on known dimensions in the image, the components can be assembled into a pipeline.

In the benchmark experiments using markers as the scale reference, the marker detection model is run in parallel with the sequential WTB and crack detection models. Where multiple markers are detected in the image, we calculate the average diagonal dimension, allowing a consensus to be reached if the bounding boxes do not perfectly describe the markers. The mm/pixel scale as calculated from the marker bounding boxes and dimensions is then used to scale the crack length from the crack bounding box. This pipeline is shown in Figure 10.

In our proposed approach of using the WTB as the scale reference, the WTB detection model and crack detection model are run sequentially, with the predicted WTB bounding box coordinates being retained for use as the scale reference and to filter out false positive crack predictions outside of the blade region. The pipeline for this method is shown in Figure 11.

Initially, crack measurement was trialled on the full WTB and global views, as well as the flat views. However, it was found that the viewing angle significantly impacted the measurement of the predicted cracks, leading to large measurement errors where the blade was viewed at an angle, whereas views close to ortho-normal showed better measurements. For this reason, the flat views were used to assess the performance on images, and ortho-normal views were used to assess performance on OMs. Crack measurements were also taken directly from the OMs for comparison, using the ruler tool in Metashape to measure the blade and crack and scaling the measurements from the blade length.

4 | Results

This section contains the results of the training and evaluation of the (i) marker, (ii) WTB and (iii) crack detection models and the results of the testing of the crack length measurement pipeline on unseen top-down images and on OMs generated from 3D reconstructions.

4.1 | Detection Models

The results of testing the Mask R-CNN models for detection of markers, WTBs and cracks are provided in Table 4. The models

were tested on the reserved test dataset, which included both flat and UAV-style views of WTBs with cracks both with and without contamination and LEE. For each model, we provide results in terms of mean precision, recall and F1 score. F1 score is the harmonic mean of precision and recall and is considered as the key metric for evaluation of the model.

Example blade, marker and crack bounding boxes are shown for Blade Types 1–3 in Figure 12.

On the reserved test images of WTB Types 1 and 2, the blade and marker detection models achieve virtually perfect performance and generalise well to images of WTB Type 3. The crack detection models show good performance on the test images of WTB Types 1 and 2, but poor performance on WTB Type 3, which is improved after fine-tuning on a small set of images of WTB Type 3.

4.2 | Crack Measurement

To determine the success of our pipeline, we present crack measurement results using both scale reference markers and WTB dimensions as the scale reference. We also calculate the error when estimating crack length based on the ground truth annotations to serve as a baseline, with error within these bounds indicating performance in line with manual inspection by a human expert. The results of the testing of the crack measurement pipeline in terms of mean absolute percentage error (MAPE) between predicted and actual crack length are shown in Table 5, comparing the scale reference objects (markers, WTBs). Measurement in OMs is also included, demonstrating the potential for application of our method to views generated from 3D reconstructions.

When trialled on images of WTB Types 1 and 2, our crack measurement pipeline showed MAPE within the range of MAPE from the annotations, with the fine-tuned crack detection model showing good results on images of WTB Type 3. In both cases, measurements based on the WTB length show slightly higher error than the markers. Our pipeline shows high MAPE when trialled on OMs, despite good performance when measuring cracks in the OMs in Metashape.

TABLE 4 | Precision, recall and F1 scores of the marker, WTB and crack detection Mask R-CNN models with test and train datasets and epochs.

Detection model	Training data	Test data	Scores			Epochs
			Precision	Recall	F1	
(i) Marker	WTB Types 1 and 2 train split	WTB Types 1 and 2 test split	0.999	0.992	0.995	15
		WTB Type 3	0.852	0.946	0.897	
WTB Types 1 and 2 test split		0.980	0.990	0.985		
WTB Type 3		0.953	1.000	0.975		
(iii) Crack		WTB Types 1 and 2 test split	0.845	0.833	0.839	
		WTB Type 3 test split	0.094	0.133	0.110	
	WTB Type 3 train split		0.705	0.470	0.564	15 + 15

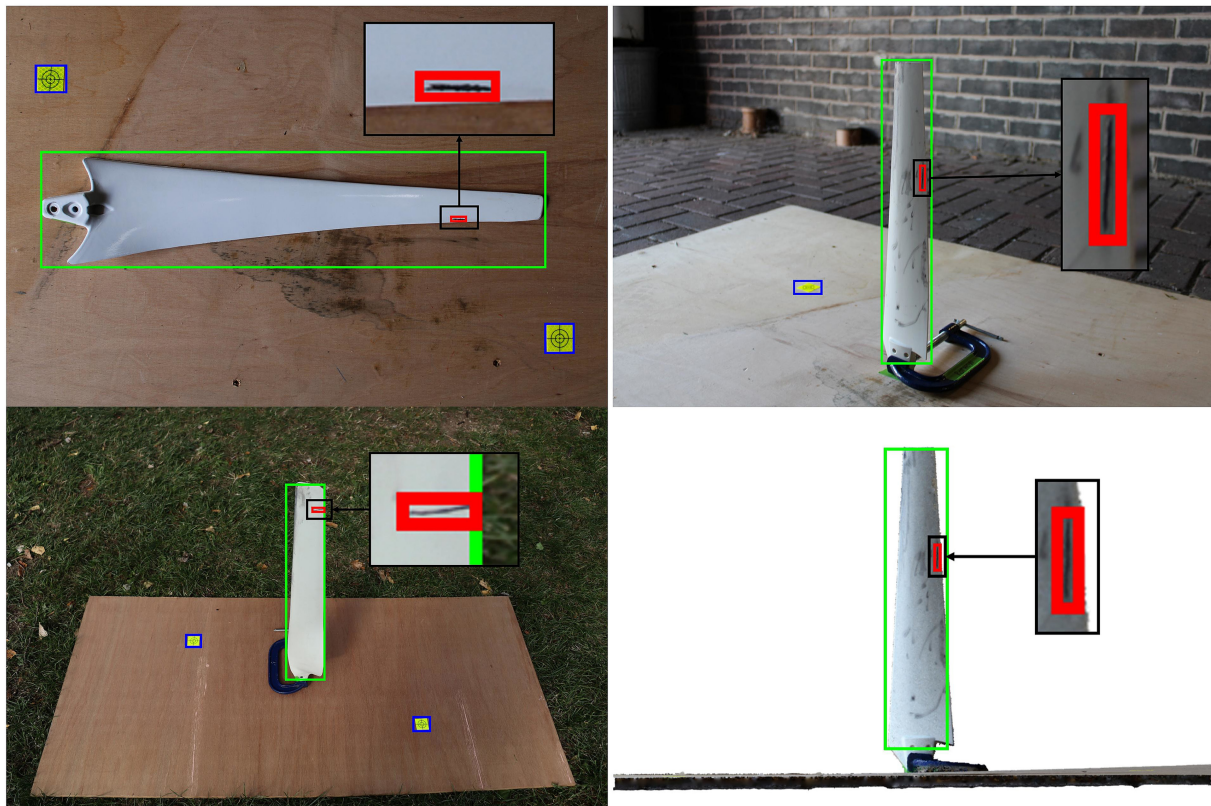


FIGURE 12 | Example images with bounding boxes showing WTBs (green), markers (blue) and cracks (red): Type 1 (top left), Type 2 (top right), Type 3 (bottom left) and an orthomosaic view of a Type 2 WTB (bottom right).

TABLE 5 | MAPE scores for measurements calculated from annotations, pipeline predictions on images, predictions on OMs and measurements from 3D reconstructions.

Measurement method	Scale reference object	MAPE
Scaling from annotations	WTB	7.507
	Marker	9.618
Our pipeline—images	WTB	5.945
	Marker	5.755
Our pipeline—images (Type 3 fine-tuned)	WTB	4.943
	Marker	4.829
Our pipeline—OMs	WTB	19.993
Metashape—OM measurement	WTB	6.070

5 | Discussion

In this section, we discuss the results of the testing of our proposed pipeline and compare our results to similar contemporary work.

In the crack detection experiments, our Mask R-CNN model showed strong performance in all metrics when trained and

tested on an 80:20 validation split of WTB Types 1 and 2, with an F1 score of 83.9%. The poor F1 score of 11% on the unseen WTB Type 3 improved to 56.4% when the model was fine-tuned on a small set of images.

Leading damage detection methods from the OSW industry show stronger performance than our model, with Li et al. [10] reporting that their SegFormer model achieves 98.5% F1 in the defect detection task across multiple types of defect, including cracks, and Hang et al. [9] reporting crack detection mAP of 97.1%. Our approach performs comparably to other WTB damage detection methods by Yu et al. [14] and Sheiati and Chen [6], which achieve F1 scores of 76.8% and 78%, respectively.

Existing concrete crack detection and measurement methods have reported high performance in detection tasks, with Li et al. [26], Nyathi et al. [25] and Giulietti et al. [27] achieving near-perfect F1 scores of 98.2%, 99.2% and 99.9%, respectively. These papers use images with cracks at a larger scale than those in our dataset and in less complex images; therefore, they are considered simpler tasks. Also, these methods use more complex damage detection frameworks, which could be further investigated in future work. However, the main contribution of this work is combining crack detection and measurement.

To evaluate our crack measurement pipeline, we compare the measurement MAPE from our pipeline (scaling on WTB length)

TABLE 6 | Comparison between our methods and existing crack measurement methods, considering image type, crack dimension and MAPE.

Methodology	Image type	Scaling method	Dimension	MAPE
Ours—from annotations				7.507
Ours—2D images	Small-scale WTBs—top down	Reference—WTB	Crack length	5.945
Ours—fine-tuned 2D images				4.943
Ours—OMs	Small-scale WTBs—OMs			19.993
Yuan et al. [23]		Reference—marker		0.8–1.6 (PE)
Minh Dang et al. [24]	Concrete cracks	Reference—brick	Length, width	10.04
Nyathi et al. [25]		Distance measurement—laser	Width	4.231
Li et al. [26]			Length	1.628

and benchmarks (scaling from annotations, scaling on markers) to results from existing works. Where published results allowed, we have calculated MAPE scores to allow comparison between cracks at different scales. The results comparing our crack measurement pipeline to our benchmarks and to other leading methods are shown in Table 6.

Our method achieves MAPE of 5.945% for crack length on WTBs, comparable to the ground truth-based measurements. This would indicate that errors present in the final model are comparable to human error and were probably introduced during the marking of the WTBs and annotation of the dataset for training the detection models. Therefore, we consider the models to match human level performance on a crack measurement task. When the crack detection model was fine tuned and tested on images of Blade Type 3, the MAPE decreased to 4.943%, indicating that fine-tuning to the intended model of WTB is beneficial both for detection and measurement.

Methods using LiDAR and laser measurement to determine the photographing distance generally achieve good results in crack measurement. Nyathi et al. [25] report their crack width measurement task achieving 0.16 mm mAE on cracks of 2–9 mm wide, with a MAPE of 4.231% calculated from their published results, outperforming our approach. Li et al. [26] report crack measurement absolute error between 0.3% and 2.8%, with MAPE of 1.628% calculated from their results. While not specifying their results, Giulietti et al. [27] report measurement error within the resolution of the laser scanner.

Reference-based approaches have shown mixed results. Use of an explicit scale reference results in low measurement error, with Yuan et al. [23] reporting measurement PE scores of 0.8%–1.6%, using a ruler for scale reference. Minh Dang et al. [24] is the most closely related approach to ours, where bricks are used to scale cracks in masonry. This work achieves measurement MAPE of 10.04%, which is greater than our method in images. Therefore, we consider our crack measurement results as positive.

Although Li et al. [10] and Hang et al. [9] report that their models achieve superior defect detection results in WTB inspection tasks, their crack measurement results cannot be compared with ours, as they do not calculate a real measurement in mm.

The majority of existing approaches are applied to relatively simple use cases, where cracks are the main feature visible, whereas our dataset contains more complex images which simulate conditions on a large-scale turbine. Largely, the cracks in the concrete and masonry use cases are directly in plane with a flat wall, photographed perpendicularly to the surface, whereas the cracks on the WTBs in our dataset are on curved surfaces, and in the UAV-style views are taken from a variety of views. We consider that the curved geometry of the WTBs creates a more complicated detection and measurement task than on flat surfaces.

Testing on OMs revealed that crack detection and measurement in composite views from 3D reconstruction is possible using our proposed pipeline, but at reduced performance compared with detection and measurement in 2D images. Overall, our approach shows higher MAPE on OMs than on images, although comparable measurement error was achieved on some examples, indicating that accurate measurement is possible under favourable conditions. The poor scores were due to a combination of missed cracks and false positive crack predictions caused by the presence of artefacts in the reconstruction. The lower MAPE of 6.07% seen in the OM measurements taken directly in Metashape indicates that there is potential to apply this pipeline to large-scale WTBs where OMs would be required to combine multiple partial WTB views and outperforms both our approach on 2D images and the similar approach proposed by Minh Dang et al. [24] using an implicit scale reference.

It is considered that application to large-scale turbine WTBs would have greater success, as the size of features increases, whereas the image quality remains comparable, increasing the potential for higher quality OMs. By building a high-quality 3D reconstruction of the WTB, orthogonal views of cracks and WTB surfaces can be generated if they are not already present, overcoming issues around measurement of cracks in UAV images. A limitation of 3D reconstruction-based measurement approaches is that time consuming pre-processing is required before measurements can be extracted, whereas our proposed approach could be adapted to run in real time using a suitable network.

6 | Conclusion

In this work, we have proposed a dedicated pipeline for crack measurement on images of WTBs using implicit scale

references—known dimensions of the WTBs. We demonstrated our proposed method on a novel dataset of small-scale WTBs with simulated crack damage and on 3D reconstructions generated using photogrammetry. We compare our approach to scaling using detected scale reference markers and to measurements calculated from human annotations, allowing comparison to related crack detection and measurement approaches and to human performance in annotating damage in images.

Overall, we find that there is strong potential for DL-based crack measurement on WTBs, although the requirements for training data and the complexity of the images remain a limitation when compared with other image-based detection tasks (e.g., COCO-style object detection) and simpler crack-measurement tasks. Our method shows good performance in 2D views and encouraging performance in views generated from 3D reconstructions. Our pipeline achieves MAPE of 5.945% using the WTB dimensions as the scale reference, comparable to the 5.755% MAPE with scale markers, and 7.507% when scaling on from ground truth annotations. Therefore, we consider our pipeline a success, without requiring additional data in terms of camera intrinsics or depth sensing and no additional pre-processing.

The positive results seen in this work justify further work in crack measurement on WTBs, both with and without explicit scale reference markers. Our approach could be extended both to other damage types on WTBs and other WT components and to other related tasks in inspection of large assets and infrastructure.

Author Contributions

Callum Rothon: conceptualisation, methodology, software, investigation, resources, data curation, writing – original draft, visualisation, project administration. **Nina Dethlefs:** conceptualisation, writing – review and editing, supervision, project administration, funding acquisition. **Robert C. Houseago:** conceptualisation, resources, writing – review and editing, supervision, project administration.

Acknowledgements

The research team was supported through the EPSRC and NERC Centre for Doctoral Training in Offshore Wind Energy and the Environment (EP/S023763/1) and the EPSRC Centre for Doctoral Training in Offshore Wind Energy Sustainability and Resilience (EP/Y034732/1), most notably through a PhD scholarship to C.R. N.D. was partially supported by the Innovation Launchpad Network AFS scheme (Grant Number EP/X528493/1).

Funding

This work was supported by the EPSRC and NERC Centre for Doctoral Training in Offshore Wind Energy and the Environment : EP/S023763/1 EPSRC Centre for Doctoral Training in Offshore Wind Energy Sustainability and Resilience : EP/Y034732/1 Innovation Launchpad Network AFS : EP/X528493/1.

Conflicts of Interest

The authors declare no conflicts of interest.

References

1. W. Wang, Y. Xue, C. He, and Y. Zhao, “Review of the Typical Damage and Damage-Detection Methods of Large Wind Turbine Blades,” *Energies* 15, no. 15 (2022): 5672, <https://doi.org/10.3390/en15155672>.
2. Bladena, Vattenfall, EON, et al., *Instruction Blade Inspections* (EUPD, 2018).
3. BVG Associates, *Guide to an Offshore Wind Farm* (BVG Associates, 2019).
4. J. Chatterjee and N. Dethlefs, “Scientometric Review of Artificial Intelligence for Operations & Maintenance of Wind Turbines: The Past, Present and Future,” *Renewable and Sustainable Energy Reviews* 144 (2021): 111051, <https://doi.org/10.1016/j.rser.2021.111051>.
5. M. Bošnjaković, M. Martinović, and K. Dokić, “Application of Artificial Intelligence in Wind Power Systems,” *Applied Sciences* 15, no. 5 (2025): 2443, <https://doi.org/10.3390/app15052443>.
6. S. Sheiati and X. Chen, “Deep Learning-Based Fatigue Damage Segmentation of Wind Turbine Blades Under Complex Dynamic Thermal Backgrounds,” *Structural Health Monitoring* 23 (2023): 539–554, <https://doi.org/10.1177/14759217231174377>.
7. X. Chen, S. Sheiati, and A. Shihavuddin, “AQUADA PLUS: Automated Damage Inspection of Cyclic-Loaded Large-Scale Composite Structures Using Thermal Imagery and Computer Vision,” *Composite Structures* 318 (2023): 117085, <https://doi.org/10.1016/j.compstruct.2023.117085>.
8. S. Sheiati, X. Jia, M. McGugan, K. Branner, and X. Chen, “Artificial Intelligence-Based Blade Identification in Operational Wind Turbines Through Similarity Analysis Aided Drone Inspection,” *Engineering Applications of Artificial Intelligence* 137 (2024): 109234, <https://doi.org/10.1016/j.engappai.2024.109234>.
9. X. Hang, X. Zhu, X. Gao, Y. Wang, and L. Liu, “Study on Crack Monitoring Method of Wind Turbine Blade Based on AI Model: Integration of Classification, Detection, Segmentation and Fault Level Evaluation,” *Renewable Energy* 224 (2024): 120152, <https://doi.org/10.1016/j.renene.2024.120152>.
10. W. Li, Z. Pan, Q. Zhu, and Y. Du, “Wind Turbine Blade Defect Detection and Measurement Technology Based on Improved SegFormer and Pixel Matching,” *Optics & Laser Technology* 179 (2024): 111381, <https://doi.org/10.1016/j.optlastec.2024.111381>.
11. J. Cuesta, U. Leturiondo, Y. Vidal, and F. Pozo, “A Review of Prognostics and Health Management in Wind Turbine Components,” *PHM Society European Conference* 8, no. 1 (2024): 1–15, <https://doi.org/10.36001/phme.2024.v8i1.4093>.
12. L. Wang and Z. Zhang, “Automatic Detection of Wind Turbine Blade Surface Cracks Based on UAV-Taken Images,” *IEEE Transactions on Industrial Electronics* 64, no. 9 (2017): 7293–7303.
13. D. Wang, Y. Zhang, and X. Yang, “Image Recognition of Wind Turbines Blade Surface Defects Based on Mask-RCNN,” in *Advanced Intelligent Technologies for Industry* (Springer, 2022), 573–584.
14. J. Yu, K. Liu, L. Qin, et al., “DMnet: A New Few-Shot Framework for Wind Turbine Surface Defect Detection,” *Machines* 10, no. 6 (2022): 487.
15. L. Zou, Y. Wang, J. Bi, and Y. Sun, “Damage Detection in Wind Turbine Blades Based on an Improved Broad Learning System Model,” *Applied Sciences* 12, no. 10 (2022): 5164.
16. C. Rothon, S. Keizer, R. Doddipatla, and N. Dethlefs, “Towards Interactive Anomaly Detection Using Natural Language,” (2024), https://bda-hull.github.io/publications/Interactive_Anomaly_Detection_using_NL_IWSDS_.pdf.
17. K. Khalili and M. Vahidnia, “Improving the Accuracy of Crack Length Measurement Using Machine Vision,” *Procedia Technology*

- 2015, no. 19 (2014): 48–55, <https://doi.org/10.1016/j.protcy.2015.02.008>.
18. L. Zhang, Z. Wang, L. Wang, Z. Zhang, X. Chen, and L. Meng, “Machine Learning-Based Real-Time Visible Fatigue Crack Growth Detection,” *Digital Communications and Networks* 7, no. 4 (2021): 551–558, <https://doi.org/10.1016/j.dcan.2021.03.003>.
19. A. Reddy, V. Indragandhi, L. Ravi, and V. Subramaniaswamy, “Detection of Cracks and Damage in Wind Turbine Blades Using Artificial Intelligence-Based Image Analytics,” *Measurement* 147 (2019): 106823.
20. A. Shihavuddin, X. Chen, V. Fedorov, et al., “Wind Turbine Surface Damage Detection by Deep Learning Aided Drone Inspection Analysis,” *Energies* 12, no. 4 (2019): 676.
21. J. Zhang, G. Cosma, and J. Watkins, “Image Enhanced Mask R-CNN: A Deep Learning Pipeline With New Evaluation Measures for Wind Turbine Blade Defect Detection and Classification,” *Journal of Imaging* 7, no. 3 (2021): 46.
22. E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, “SegFormer: Simple and Efficient Design for Semantic Segmentation With Transformers,” (2021).
23. Y. Yuan, Z. Ge, X. Su, et al., “Crack Length Measurement Using Convolutional Neural Networks and Image Processing,” *Sensors* 21, no. 17 (2021), <https://doi.org/10.3390/s21175894>.
24. L. Minh Dang, H. Wang, Y. Li, et al., “Deep Learning-Based Masonry Crack Segmentation and Real-Life Crack Length Measurement,” *Construction and Building Materials* 359 (2022): 129438, <https://doi.org/10.1016/j.conbuildmat.2022.129438>.
25. M. A. Nyathi, J. Bai, and I. D. Wilson, “Deep Learning for Concrete Crack Detection and Measurement,” *Metro* 4, no. 1 (2024): 66–81, <https://doi.org/10.3390/metrology4010005>.
26. Y. Li, H. Zhao, Y. Wei, et al., “Vision-Guided Crack Identification and Size Quantification Framework for Dam Underwater Concrete Structures,” *Structural Health Monitoring* 24, no. 4 (2024): 2125–2148, <https://doi.org/10.1177/14759217241287906>.
27. N. Giulietti, G. M. Revel, and P. Chiariotti, “Automated Vision-Based Concrete Crack Measurement System,” *Measurement* 242 (2025): 115858, <https://doi.org/10.1016/j.measurement.2024.115858>.
28. J. Sterckx, M. Vlamincx, K. De Bauw, and H. Luong, “Accurate and Robust 3D Reconstruction of Wind Turbine Blade Leading Edges From High-Resolution Images,” *Automation in Construction* 175 (2025): 106153, <https://doi.org/10.1016/j.autcon.2025.106153>.
29. S. Pierce, K. Burnham, L. McDonald, et al., “Quantitative Inspection of Wind Turbine Blades Using UAV Deployed Photogrammetry,” in *9th European Workshop on Structural Health Monitoring (EWHM 2018)* (NDT.net, 2018), 1–12, <https://www.ndt.net/article/ewshm2018/papers/0195-Pierce.pdf>.
30. A. K. V. de Oliveira, M. K. Bracht, M. Aghaei, and R. R  ther, “Automatic Fault Detection of Utility-Scale Photovoltaic Solar Generators Applying Aerial Infrared Thermography and Orthomosaicking,” *Solar Energy* 252 (2023): 272–283, <https://doi.org/10.1016/j.solener.2023.01.058>.
31. F. Lu, K. Li, Y. Nie, et al., “Object Detection of UAV Images From Orthographic Perspective Based on Improved YOLOv5s,” *Sustainability* 15, no. 19 (2023), <https://doi.org/10.3390/su151914564>.
32. A. Shihavuddin and X. Chen, *DTU-Drone Inspection Images of Wind Turbine* (Technical University of Denmark (DTU), 2018).
33. C. Yang, X. Liu, H. Zhou, Y. Ke, and J. See, “Towards Accurate Image Stitching for Drone-Based Wind Turbine Blade Inspection,” *Renewable Energy* 203 (2023): 267–279.
34. T. Woelk and A. Sier, “Generating High Quality Outputs From Drone Imagery,” (2025).
35. S. Ullman, “The Interpretation of Structure From Motion,” *Proceedings of the Royal Society of London Series B* 203, no. 1153 (1979): 405–426, <https://doi.org/10.1098/rspb.1979.0006>.
36. D. Li, D. Song, S. Liu, et al., “Camera Pose Estimation Based on Global Structure From Motion,” *Multimedia Tools and Applications* 79 (2020): 23223–23242, <https://doi.org/10.1007/s11042-020-09045-8>.
37. DJI Enterprise, “A Close Look at Drone Orthomosaics in Surveying and How to Get the Most From Your Data,” (2021).
38. JOUAV, “Orthomosaic Mapping: What’s Orthomosaic & How to Make It?” (2025).
39. Agisoft LLC, *Agisoft Metashape* (Agisoft LLC, 2025).
40. R. Girshick, J. Donahue, T. Darrell, and J. Malik, “Rich Feature Hierarchies for Accurate Object Detection and Semantic Segmentation,” (2014).
41. R. Girshick, “Fast R-CNN,” (2015).
42. S. Ren, K. He, R. Girshick, and J. Sun, “Faster R-CNN: Towards Real-Time Object Detection With Region Proposal Networks,” (2015).
43. K. He, G. Gkioxari, P. Doll  r, and R. B. Girshick, “Mask R-CNN,” (CoRR, 2017); abs/1703.06870.
44. J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You Only Look Once: Unified, Real-Time Object Detection,” (2015).
45. O. Russakovsky, J. Deng, H. Su, et al., “ImageNet Large Scale Visual Recognition Challenge,” (2014).
46. T. Y. Lin, M. Maire, S. Belongie, et al., “Microsoft COCO: Common Objects in Context,” (2015).
47. Z. Xin, S. Chen, T. Wu, Y. Shao, W. Ding, and X. You, “Few-Shot Object Detection: Research Advances and Challenges,” (2024).
48. A. Parnami and M. Lee, “Learning From Few Examples: A Summary of Approaches to Few-Shot Learning,” (2022).
49. F. Zhuang, Z. Qi, K. Duan, et al., “A Comprehensive Survey on Transfer Learning,” (2020).
50. A. Dutta and A. Zisserman, “The VIA Annotation Software for Images, Audio and Video,” in *Proceedings of the 27th ACM International Conference on Multimedia* (ACM, 2019).